

$$\sum_{i=1}^n (u^t y_i)^2 = \sum_{i=1}^n u^t y_i y_i^t u \quad | \quad 25/10/2025$$

$$= u^t \left(\sum_{i=1}^n y_i y_i^t \right) u = u^t S u$$

$$\hat{C} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^t$$

$$u = \sum_{j=1}^d a_j \underbrace{v_j}_{\substack{\text{scalar} \\ \text{d} \times 1}} \quad \begin{array}{l} \text{d} \times 1 \\ \downarrow \\ \text{d} \times 1 \end{array}$$

Variance of data along direction u

$$Z_i = u^t (x_i - \bar{x}) \quad \frac{1}{n} \sum_{i=1}^n Z_i = 0$$

$$\text{Var}(Z_i) = \frac{1}{n} \sum_{i=1}^n (u^t (x_i - \bar{x}))^2$$

$$= \frac{1}{n} \sum_{i=1}^n (u^t y_i)^2 = \frac{1}{n} \sum_{i=1}^n u^t y_i y_i^t u$$

$$= \frac{1}{n} u^t \left(\sum_{i=1}^n y_i y_i^t \right) u = \frac{1}{n} u^t S u$$

If $u = v_1$ is eigenvector corresp. to largest eigenvalue, then $u^t S u = u^t \lambda_1 u = \lambda_1 u^t u = \lambda_1$